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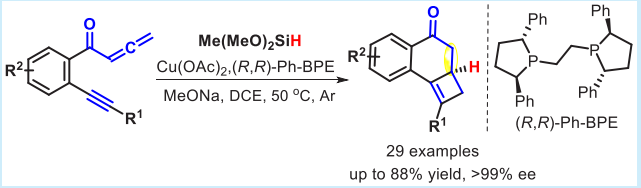
Enantio- and Regioselective CuH-Catalyzed Conjugate Reduction of Yne−Allenones

Jia-Yin Wang, Guigen Li,* Wen-Juan Hao, and Bo Jiang*

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ABSTRACT: A new asymmetric catalytic conjugate reduction of yne−allenones to synthesize enantioenriched cyclobuta[*a*]-naphthalen-4(2*H*)-ones has been established that uses copper−bisphosphine complexes as catalysts and gives excellent regio- and enantioselectivities (≥99% ee) in most cases. This protocol tolerates a broad scope of substrates, exhibits high compatibility with various substituents, and gives excellent stereoselectivity, providing a catalytic and efficient entry to fabrication of synthetically important chiral 6−6−4 tricarbocyclic scaffolds.



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Over the past decades, the asymmetric transfer hydrogenation reaction has gradually matured and become one of the premier methods to install stereocenters in organic molecules. Most importantly, several industrial processes rely on the chemical prowess of this transformation due to its high efficiency and environmental friendliness.[1] Generally, transition-metal-catalyzed asymmetric hydrogenations of *α,β*-unsaturated substrates, including *α,β*-unsaturated carboxylic acids, esters, amides, ketones, aryl alkenes, aliphatic alkenes, and even alkynes, have been developed with considerable success.[2] Among them, the use of hydrogen gas as the hydrogen source is among the most significant strategies to achieve asymmetric hydrogenation, and various successful examples have been reported with excellent results by many research groups.[3] However, this method relies on noble metals as catalysts, including Rh, Ru, Ir, and Pd, which are highly expensive and pose significant threats to the environment and public health even at low catalyst loadings. Compared with the above transition metals, other earth-abundant transition metals, such as Mn,[4] Fe,[5] Co,[6] Ni,[7] and Cu,[8] have also been proven to be efficient in asymmetric catalytic hydrogenation with the benefits of lower costs and comparatively lower toxicity, thus providing greener and more sustainable prospects for industrial production of chiral products.

On the basis of expenses of metals, mild catalytic conditions, and operational simplicity, the use of asymmetric copper hydride (CuH) catalysis as the reducing species in asymmetric hydrofunctionalization has been paid more attention and has emerged as powerful and versatile synthetic means in regard to the formation of valuable chiral compounds during the past two decades.[9] Particularly, the asymmetric 1,4-reduction of *α,β*-unsaturated Michael acceptors catalyzed by CuH species[10] has been well-developed. The Buchwald group pioneered an enantioselective 1,4-conjugate reduction via in situ generation of chiral CuH species by using CuCl as the copper source, (*S*)-*p*-*tol*-BINAP as the chiral ligand, and poly(methylhydrosiloxane) (PMHS) as the hydrogen source. In this strategy, enones, unsaturated esters, lactones, and lactams undergo 1,4-conjugate asymmetric reduction to afford the corresponding chiral products in good yields with excellent enantioselectivities (Scheme 1a).[11] Subsequently, Lipshutz and co-workers used (*R*)-DTBM-SEGPHOS-derived chiral CuH species to establish enantioselective 1,4-reduction of PMHS onto *β*-substituted acyclic enones, leading to chiral products in high yields with high enantioselectivities.[12] Despite these great advances in this field, an asymmetric 1,4-reduction reaction for in situ generation of fused tricyclic enones has not been reported to date.

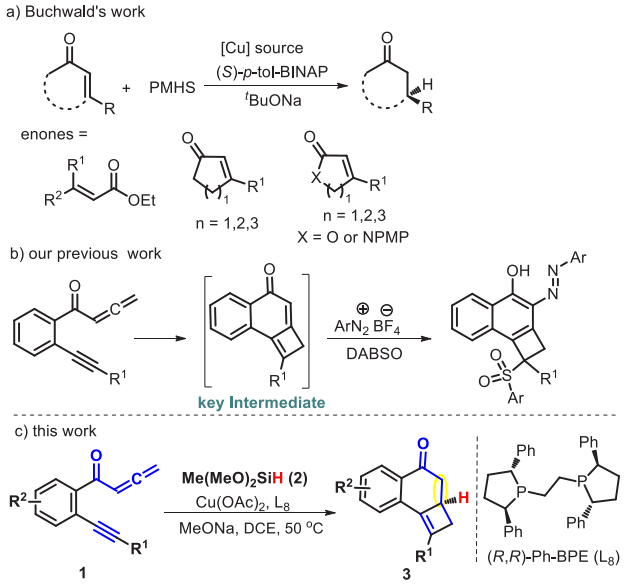
We recently developed a [2 + 2] cycloaddition/radical-induced 1,4-conjugate addition cascade by using yne−allenones, aryldiazonium salts, and DABCO-bis(sulfur dioxide) (DABSO), leading to the formation of cyclobuta[*a*]-naphthalen-4-ols.[13] During this process, the [2 + 2] cycloadduct as a 1,4-addition radical acceptor acted as a key intermediate. To continue our efforts on this project[14] and asymmetric catalysis,[15] we became interested in the design of a new CuH-catalyzed asymmetric conjugate reduction strategy that takes advantage of the methodology invented by Buchwald's group.[11] Herein we report the first asymmetric conjugate reduction of yne−allenones as an approach to regioselectively afford a wide variety of cyclobuta[*a*]-naphthalen-4(2*H*)-one derivatives through CuH-catalyzed

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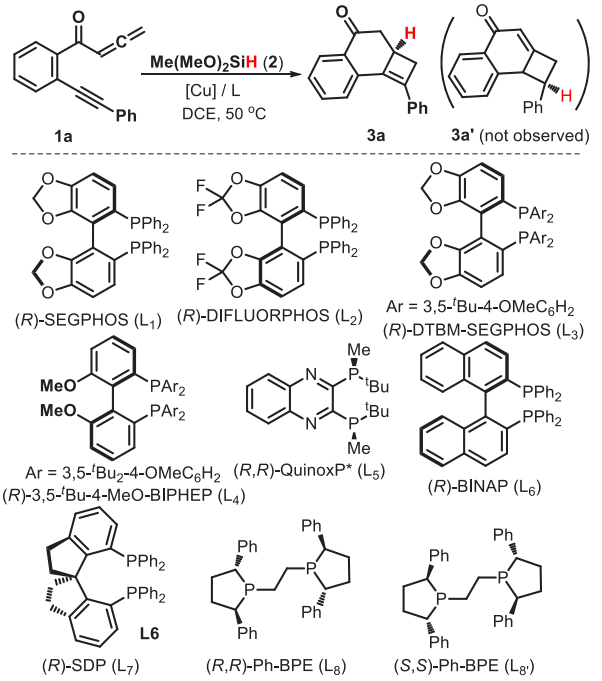
Scheme 1. Profiles of Asymmetric Conjugate Reduction



enantioselective 1,4-conjugate reduction utilizing dimethoxy(methyl)silane (DMMS, **2**) as a readily available hydrogen source.

We initiated our research on the identification of effective chiral copper catalysts for the reaction between the benzene-linked yne−allenone **1a** with **2**. The chiral CuH complexes generated in situ from $Cu(OAc)_2$ alone with a series of chiral bisphosphine ligands were screened, and the results are shown in Table 1 (entries 1−8). At the beginning, the reaction was performed in the presence of 10 mol % $Cu(OAc)_2$ and 10 mol % *R*-SEGPHOS ($\mathbf{L_1}$) in DCE at 50 °C for 12 h, and the desired cyclobuta[*a*]naphthalene-4(2*H*)-one product **3a** was obtained with 93% ee, albeit in a very low yield (10%) (entry 1). Interestingly, the 1,6-adduct **3a′** was not observed at all,[13,14] demonstrating the complete regioselectivity of this asymmetric reaction. The regioselectivity arise because chelation of the carbonyl group from the cyclohexone to the CuH catalyst benefits the proximal 1,4-addition and is not conducive to the distal 1,6-addition. Replacing $\mathbf{L_1}$ with *R*-DIFLUORPHOS ($\mathbf{L_2}$) or *R*-DTBM-SEGPHOS ($\mathbf{L_3}$) resulted in inferior results with regard to the yield (entries 2 and 3). The use of $\mathbf{L_4}$ or $\mathbf{L_5}$ afforded both lower yield and lower enantioselectivity (entries 4 and 5). Both (*R*)-BINAP ($\mathbf{L_6}$) and (*R*)-SDP ($\mathbf{L_7}$) completely suppressed the reaction process (entries 6 and 7). Interestingly, the presence of (*R*,*R*)-Ph-BPE ($\mathbf{L_8}$) improved both the chemical yield and the enantioselectivity (43% yield, >99% ee; entry 8). We found that involving appropriate bases could increase the reaction yield with simultaneous retention of outstanding enantioselection (entries 9−16). For example, the use of 1.5 equiv of LiOH afforded the product **3a** in 60% yield with >99% ee (entry 10), and other bases such as KOH, ^{t}BuOLi, MeONa, and KOAc also benefited the reaction process (entries 12−15), with MeONa giving the best outcome in regard to the yield and enantioselectivity (entry 14). Other Cu salts, including Cu(OAc), $CuCl_2$, and $Cu(acac)_2$, were then tested together with (*R*,*R*)-Ph-BPE in this transformation, and all afforded unsatisfactory results (entries 16−18). Subsequently, different silanes were evaluated, including Ph_2SiH_2, $PhSiH_3$, and Ph_3SiH (entries 19−21). The reaction of **1a** with Ph_2SiH_2 delivered **3a**

Table 1. Optimization of the Reaction Conditions[a]



entry	Cu cat.	L	base (equiv)	yield (%)[b]	ee (%)[c]
1	$Cu(OAc)_2$	$\mathbf{L_1}$	–	10	93
2	$Cu(OAc)_2$	$\mathbf{L_2}$	–	23	96
3	$Cu(OAc)_2$	$\mathbf{L_3}$	–	33	90
4	$Cu(OAc)_2$	$\mathbf{L_4}$	–	20	79
5	$Cu(OAc)_2$	$\mathbf{L_5}$	–	23	80
6	$Cu(OAc)_2$	$\mathbf{L_6}$	–	trace	–
7	$Cu(OAc)_2$	$\mathbf{L_7}$	–	trace	–
8	$Cu(OAc)_2$	$\mathbf{L_8}$	–	43	>99
9	$Cu(OAc)_2$	$\mathbf{L_8}$	LiOH (0.5)	48	>99
10	$Cu(OAc)_2$	$\mathbf{L_8}$	LiOH (1.5)	60	>99
11	$Cu(OAc)_2$	$\mathbf{L_8}$	LiOH (2.0)	57	>99
12	$Cu(OAc)_2$	$\mathbf{L_8}$	KOH (1.5)	46	99
13	$Cu(OAc)_2$	$\mathbf{L_8}$	^{t}BuOLi (1.5)	68	>99
14	$Cu(OAc)_2$	$\mathbf{L_8}$	MeONa (1.5)	75	>99
15	$Cu(OAc)_2$	$\mathbf{L_8}$	KOAc (1.5)	69	>99
16	Cu(OAc)	$\mathbf{L_8}$	MeONa (1.5)	37	>99
17	$CuCl_2$	$\mathbf{L_8}$	MeONa (1.5)	23	97
18	$Cu(acac)_2$	$\mathbf{L_8}$	MeONa (1.5)	57	>99
19[d]	$Cu(OAc)_2$	$\mathbf{L_8}$	MeONa (1.5)	54	>99
20[e]	$Cu(OAc)_2$	$\mathbf{L_8}$	MeONa (1.5)	nd[g]	–
21[f]	$Cu(OAc)_2$	$\mathbf{L_8}$	MeONa (1.5)	trace	–
22	$Cu(OAc)_2$	$\mathbf{L_{8'}}$	MeONa (1.5)	72	> −99

[a]Reaction conditions: **1a** (0.1 mmol), **2** (0.3 mmol), [Cu] (10 mol %), ligand (10 mol %), base (*y* equiv), solvent (1.0 mL), 50 °C, 12 h, under Ar. [b]Isolated yields based on substrate **1a**. [c]Determined by chiral HPLC analysis. [d]Ph_2SiH_2 was used. [e]$PhSiH_3$ was used. [f]Ph_3SiH was used. [g]Not detected.

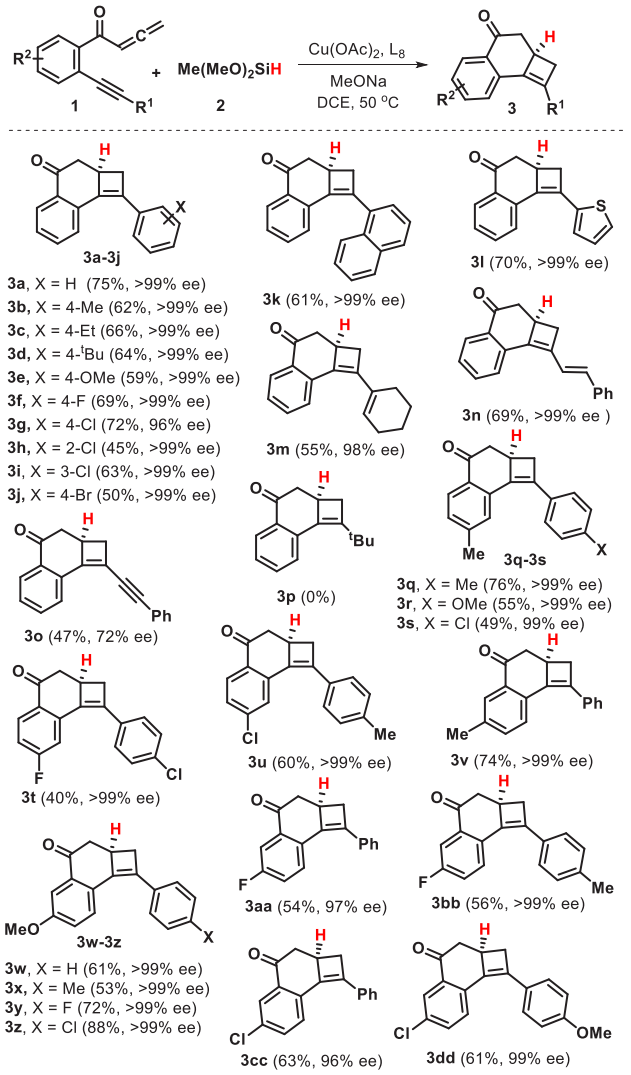
in moderate yield with good enantiocontrol (entry 19). In contrast, the latter two did not give the desired product **3a** (entries 20 and 21). It is noteworthy that the opposite configuration ligand of (*S*,*S*)-Ph-BPE ($\mathbf{L_{8'}}$) was also examined for this reaction and afforded **3a** in 72% yield, but the opposite enantioconfiguration was observed (ee > −99% ee) (entry 22).

Under the optimized conditions, the scope of this [2 + 2] cycloaddition/asymmetric 1,4-reduction reaction was investigated by utilizing a variety of preformed yne−allenones **1**

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(Scheme 2). A great number of cyclobuta[*a*]naphthalen-4(2*H*)-ones **3a**–**cc** were furnished in modest to good yields

Scheme 2. Substrate Scope of Yne–Allenones[a]

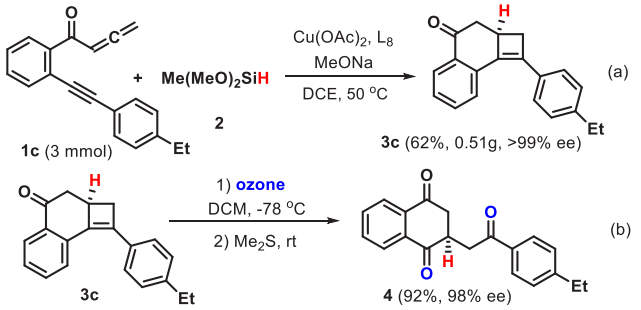


[a]Reaction conditions: **1** (0.2 mmol), **2** (0.6 mmol), $Cu(OAc)_2$ (10 mol %), (*R*,*R*)-Ph-BPE (10 mol %), MeONa (1.5 equiv), DCE (2.0 mL), 50 °C, 12 h, under Ar. Isolated yields based on substrates **1** are shown. The ee values were determined by chiral HPLC analysis.

(40–88%) with generally high enantioselectivities (72 to >99% ee). Both electron-donating (methyl (**1b**), ethyl (**1c**), *tert*-butyl (**1d**), or methoxy (**1e**)) and electron-withdrawing (fluoride (**1f**), chloride (**1g**–**i**), or bromide (**1j**)) groups at different positions of the arylalkynyl motif (R^1) tolerated this reductive reaction system, leading to the corresponding products **3b**–**j** in 50–75% yield with 96 to >99% ee. In addition, the sterically encumbered 1-naphthyl (1-Np) analogue **1k** was proven to be a suitable reactant, allowing a similar 1,4-reduction process toward the desired product **3k** in 61% yield with >99% ee. Similarly, the 2-thienyl analogue **1l** showed high reactivity, giving the corresponding 2-thienyl-substituted product **3l** in 70% yield with >99% ee. Interestingly, cyclohexenyl-substituted yne–allenone **1m** was effective for accessing the corresponding enantioenriched product **3m** with comparable efficiency. Furthermore, styryl- and phenylethynyl-substituted counterparts (**1n** and **1o**) were adopted to demonstrate the compatibility of this protocol and furnished the target products **3n** and **3o**, respectively, although a relatively lower yield and enantioselectivity were obtained in the latter case. Unfortunately, *tert*-butyl-substituted yne–allenone **1p** resulted in severe decomposition without forming any of the corresponding product. Alternatively, substituents with electron-efficient and -deficient natures at C4 or C5 of the internal arene ring were successfully employed in the present asymmetric reaction. Various substituents, including methyl, methoxy, fluoro, and chloro, enabled CuH-catalyzed asymmetric conjugate reduction to afford the chiral cyclobuta[*a*]naphthalen-4(2*H*)-ones **3q**–**dd** in moderate to good yields with excellent enantioselectivities (96 to >99% ee). The absolute configuration of (*R*)-**3c** was unambiguously determined by X-ray crystallographic analysis (CCDC 2064795), and those of its derivatives were assigned by analogy (see the SI).

The significant potential of this protocol was demonstrated by carrying out the amplification experiment for the synthesis of (*R*)-**3c** with retention of excellent enantioselectivity (Scheme 3a). Because chiral quinones with quaternary

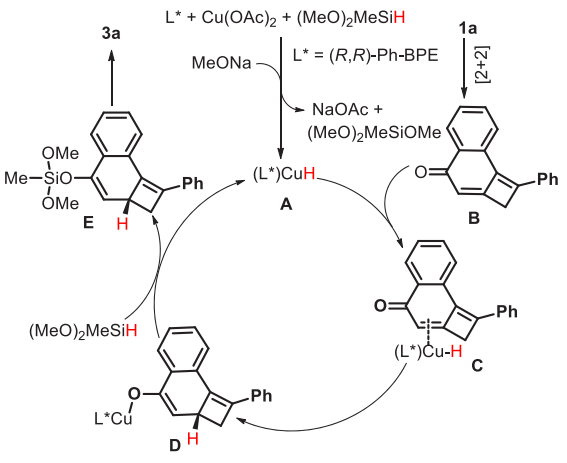
Scheme 3. Scale-Up Experiment and Derivatization Reaction



stereocenters have been found to show interesting biological properties,[16] ozonation of (*R*)-**3c** followed by reductive workup with Me_2S was conducted, affording (*R*)-2,3-dihydronaphthalene-1,4-dione **4** in 92% yield without loss of ee (Scheme 3b).

On the basis of the experiments and previous work on the CuH-catalyzed asymmetric conjugate reduction reaction,[11,12] our proposed reaction pathway is illustrated in Scheme 4. The

Scheme 4. Possible Mechanism for the Formation of 3



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catalytic L*CuH species **A** is initially generated via the reaction of $Cu(OAc)_2$ with (*R*,*R*)-Ph-BPE and a stoichiometric amount of the hydrosilane.[17] Copper hydride species **A** reacts with Michael acceptor **B** generated from intramolecular [2 + 2] cycloaddition of yne−allenone **1a**, leading to π complex **C**. Delivery of the hydride to the β-carbon atom gives copper enolate intermediate **D**, which undergoes σ-bond metathesis with the hydrosilane, followed by closure of the cycle by regeneration of the catalytic Cu−H species **A** and release of species **E**, which is converted into product **3a** through tautomerization after workup.[9c,18]

In summary, we have successfully developed an unprecedented [2 + 2] cycloaddition/asymmetric 1,4-conjugate reduction of yne−allenones via a regioselective CuH-catalyzed conjugate reduction reaction. This reaction provides efficient and straightforward access to a variety of chiral cyclobuta[*a*]-naphthalen-4(2*H*)-ones in acceptable yields with high enantioselectivity. Further studies on expanding the substrate scope and developing other types of enantioselective transformations by employing CuH species are being conducted in our laboratory.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.1c00892.

Experimental procedures and characterization data for all products, including NMR spectra (PDF)

Accession Codes

CCDC 2064795 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, U.K.; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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